

Semistable reductions

[Sch] Variation of Hodge str.

Our Goal

[K+71]

[KMP71]

Thm (Fujita's 2nd Thm)

[Cat08]

Let $f: X \rightarrow B$ a fibration of a ^{scat 08} cpt Kähler mfd X over a projective curve B , and consider the direct image (of the relative dualizing) sheaf $V := f_* \omega_{X/B}$. Then V splits as a direct sum $V = A \oplus Q$, where A is an ample vector bundle and Q is a unitary flat bundle.

Aim of this part is to show that $f_*(\mathcal{O}_X(K_X - f^*K_B))$ it suffices to prove the theorem in the semistable case, i.e. where each fibre is reduced and a normal crossing divisor.

Q: why is this situation good? ^A the local monodromy = unipotent!

Variations of Hodge Structure X, Y : sm proj var. $f: X \rightarrow Y$ morph. $B \subset Y$: SNC $R := f^{-1}(B)$

$$\begin{array}{ccc} f: & X & \longrightarrow Y \\ & \cup & \\ & R & \longrightarrow B \end{array}$$

We assume $f: X \setminus R \rightarrow Y \setminus B$ is sm.

cohomology groups of the fibers $X_y := f^{-1}(y)$ give local systems $R^i f_* \mathbb{C}_{X \setminus R}$ over $Y \setminus B$ which are variations of HS.

Set $n = \dim X - \dim Y$. The identification

$$H^i(X_y, \omega_{X_y}) \cong H^{n-i}(X_y) \subset H^{n+i}(X_y, \mathbb{C})$$

give sub vect bund.

$$R^i f_* \omega_{X/Y}|_{Y \setminus B} \hookrightarrow \mathcal{O}_{Y \setminus B} \otimes_{\mathbb{C}} R^{n+i} f_* \mathbb{C}|_{Y \setminus B}.$$

We get flat metric on local system. (last time)

(Borel cf [Sch 73, 4.5])

[2]

Let $\gamma_j \subset Y \setminus B$ a small loop around an irred comp $B_j \subset B$ with base point y_j . as we move around γ_j , the resulting linear map $P_j \in GL(H^{ht, i}(X_{y_j}, \mathbb{C}))$ (called the local monodromy around B_j) is quasi-unipotent. (all eigenvalues of P are roots of unity)
(the power of P_j is unipotent)

A key technical point is that as long as Y is sm and B is a snc divisor, we can ignore what happens over any codim 2 subset $Z \subset Y$.

Thm* (Katz cf [Kat 74])

Notation as above, Assume that there is a codimension 2 subset $Z \subset Y$ such that $T^*(B \setminus Z)$ is a reduced snc divisor, then $R^i \Gamma_* \mathbb{C}_X / Y \setminus B$ has unipotent monodromies for every i

Idea If the monodromy $P_j \in GL(H^i(X_{y_j}, \mathbb{C}))$ is not unipotent but $P_j^{m_j}$ is unipotent. then we can create unipotent monodromies by a simple base change.

Prp Notation above, Let Y' be a smooth var. and $\pi: Y' \rightarrow Y$ a finite Galois map s.t m_j divides the ramification order of π above B_j . $\pi^{-1}(Z) \subset Z$
Let $X' \rightarrow X \times_Y Y'$ be any resolution. Then $R^i \Gamma_* \mathbb{C}_X / Y \setminus \pi^{-1}(B)$ has unipotent monodromies

The semi-stable reduction Thm of Mumford says that the assumption of Thm* can be satisfied

after suitable base change $Z: Y' \rightarrow Y$.

[3]

Semistable reduction (see [KMSD73]
(for curves) or [Stack project])

Setting $f: X \rightarrow B$ fibration of cpt Kähler and X
over proj curve B .

There exists a cyclic Galois covering of B ,
 $B' \rightarrow B = B'/G$ s.t. the normalization X'' of the
fibre product $B' \times_B X$ admits a resolution $X' \rightarrow X''$
s.t. the resulting fibration $f': X' \rightarrow B'$ has all the fibres
which are reduced and normal crossing divisors.

$$\begin{array}{ccccc} X' & \xrightarrow{v'} & X'' & \xrightarrow{v''} & X \\ f' \downarrow & & \downarrow & & \downarrow f \\ B' & \xrightarrow{\text{Id}} & B' & \xrightarrow{u} & B \end{array}$$

Prmk At each singular fiber $F = \sum m_i C_i$

corresponding to a point $t=0$ on B
the thm yields a base change $t = z^n$ where
 $m_i/n \in \mathbb{Z}$ and $n \gg 0$

Rmk on semistable reduction

- It was already known for the case of Abelian varieties
by Gmthendieck and others.

- The first proof of the semistable reduction thm
for curves by Deligne and Mumford. (1969)

"The irreducibility of the space
of curves of given genus"

This proves ^{the} thm by reducing the problem to
the case of Abelian varieties. (Jacobian Var).

- The sheaf $V' = t'_* \omega_{X'/B'}$ is a subsheaf of the sheaf $u^*(V)$ where $V = t_* \omega_X/B$.
- and the cokernel $u^*(V)/V'$ is concentrated on the set of points corresponding to singular fibres of f .
- In particular, since V and V' are semipositive by Fujita's first thm, (chf)

(each quot bundle has degree $\deg Q \geq 0$)

if V' satisfies the property

that for each degree 0 quotient bundle Q' of V' then there is a splitting $V' = E' \oplus Q'$ for the projection $p: V' \rightarrow Q'$ and Q' unitary flat.

Then V' splits as the direct sum $V' = A \oplus Q$ where A is an ample vector bundle and Q is a flat unitary bundle, and the same conclusion holds also for V .

This prop suggests us in order to prove Fujita 2nd Thm, it suffices to prove the Thm in the semi-stable case.

Idea of proof

- It suffices to work locally around each point $P' \in B'$ mapping to a point P corresponding to a singular fibre F of f and consider the base change $t = \mathbb{C}^n$ where n may be assumed not to depend on the point P . By the Hurwitz formula.

$$K_{B'} = u^*(K_B) + \sum_{P'} (n-1)P', \quad K_{X'} = v^*K_X + R$$

$$\Rightarrow \omega_{X'/B'} = K_{X'} - f'^*(K_{B'}) = v^*(\omega_{X/B}) - \underbrace{\left(\sum_{P'} (n-1)F' - R \right)}_{\text{supported on the inverse images of the singular fibres of } f}$$

① is effective

For the 2nd assertion

using the similar discussion with Prop 2.3 step 3~5.

Harder-Narasimhan filtration of V

$$\leadsto 0 \rightarrow U \rightarrow V \rightarrow Z \rightarrow 0 \quad \text{where } U: \text{ ample} \\ \deg(Z) = 0.$$

We set $Q := u^*(Z)$

$$\text{observe } V' \twoheadrightarrow Q \leadsto V' \stackrel{\exists}{=} E' \oplus Q$$

claim this $Q \subset V'$ induces a splitting $V = U \oplus Z$.

$$\textcircled{1} \quad V = u_* (u^* V)^G$$

homomorphism

$\Rightarrow V' \rightarrow u^* V$ induces a chain of homomorphism.

$$Z = u_* (u^* Z)^G \rightarrow u_* (u^* Z)^G \oplus u_* (A)^G = u_* (V')^G$$

whose composition is the identity. $\diamond \rightarrow V \rightarrow Z$

③ last thing to show is Z is flat.

See paper.

(idea): use orbifold fundamental group

see [Cat 08]

Ret

[CD]: Catanese, Pettnweiler Answer to a Question
by Fujita on variation of Hodge
structures

Ch 2.2.

[Sch]: ^{Schmid} Variation of Hodge structures ^{Lemmas} ~~Ch 4.5~~.

[Kat] Katz. The regularity theorem in algebraic geometry
§ VII

unusually

[KMSD] Kempf, Knudsen, Mumford Saint. Donat.
semi-st. red. Toroidal embeddings. I ch 2. by Mumford.

[Cat] Catanese. Differentiable and deformation
type of algebraic surfaces.
real and symplectic structures.

orbitoid fund.
grp ch 5.1

~~[ACF]~~ kollár Kodaira's canonical bundle formula
and subadjunction.

§ 8.9

ch 8 in "Flips for 3-folds and
Corti et al 4-folds"